Form AR 140

The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT509
Subject Title	Advanced Textiles Materials and Technologies
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject provides a high level of understanding in modern textiles materials and the latest technologies used in the global fashion industry. Students will also be able to gain in-depth understanding of key factors shaping the design and selection of innovative materials used in advanced engineering applications, their processing, properties and stability.
Intended Learning Outcomes *(Note 1)*	Upon completion of the subject, students will be able to: a) Analyze the modern production skills and techniques for textile and garment production; b) Describe and identify the interfaces between different materials, quality standards, and price levels for appropriate applications; c) Resolve the technical and quality problems during the textile production process for possible improvements; and d) Identify the latest market trends and technology development in the fashion and textile industry and analyze the required facilities and activities for fashion global sourcing and supply chain network.
Subject Synopsis/ Indicative Syllabus *(Note 2)*	1. <u>Textiles Innovation and Technologies</u> Conventional and new innovative materials; diverse and novel applications for textile materials; digital technology and development of new equipment, artefacts and devices. 2. <u>Design, Innovations and Sustainability</u> Innovative materials and processes; ethical and environmental impact of textile design and production; sustainability issues and solutions in textile production.

3. <u>Materials Performance and Analysis</u>
   Quality assessment for new innovative textile products; industry standard and evaluation; industry testing and processing; recent research and development of textiles materials.

4. <u>Manufacturing Processes and Technologies</u>
   Innovation in product development; theoretical and practical methods for product analysis.

Teaching/Learning Methodology

(Note 3)

This subject will be delivered through lectures, demonstrations, laboratory and trade fair visits to introduce knowledge, skills and procedures for various kinds of textile materials, with emphasis on developing students' analytical and cognitive skills in the latest textile design and production.

Assessment Methods in Alignment with Intended Learning Outcomes

(Note 4)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Course work					
(a) individual project	20%	✓	✓		✓
(b) group project	20%	✓	✓	✓	✓
(c) presentation	10%	✓	✓	✓	
2. Examination	50%	✓	✓	✓	
Total	100 %				

Continuous assessments with group and individual projects are designed to allow students the opportunity for theoretical application in real case study, further, organizational skills and leadership skills are enhanced.

The examination assess students' understanding and applications of different advanced materials and technologies, and problem-solving ability in resolving quality and technical issues during the production process.

Student Study Effort Expected

Class contact:	
▪ Lectures	10 Hrs.
▪ Tutorial, presentation and critics	15 Hrs.
▪ Studio/laboratory	14 Hrs.

	Other student study effort:	
	▪ Assignments	36 Hrs.
	▪ Self-study	30 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Books Hayes, S., Venkatraman, P. (2017) *Materials and Technology for Sportswear and Performance Apparel*, London: Routledge Tate, D. (2018) *Textile Engineering: Materials, Design and Technology*, New York: NY Research Press Ul-Islam, S. (2019) *The impact and prospects of green chemistry for textile technology*, Cambridge: Woodhead Publishing Fawcett, I., Tranter, D., Treuherz, P. (2018) *Design and technology: Textile-based materials,* London: Hodder Education Franklin K., Till C. (2019) *Radical Matter: Rethinking Materials for a Sustainable Future.* London: Thames and Hudson Nayak, R. (2019) *Sustainable Technologies for Fashion and Textiles.* Cambridge: Woodhead. Journals Apparel International International Journal of Clothing Science and Technology International Textile Bulletin (Fabric Forming) J.S.N. International Journal of the Textile Institute Knitting International Melliand Textiberichte (Melliand English) Textile Asia Textile Institute and Industry Textile Praxis International Textile Research Journal Internet Resources www.wgsn.com www.businessoffashion.com/ www.voguebusiness.com/ https://wwd.com https://viewzines.com	

Note 1: Intended Learning Outcomes
Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology
This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method
This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.